

نام: علی قاضی
کتاب: داتا سٹرکچر

تمرین مربوط به الگوریتم Floyd

$$D_0 = W = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 5 & \infty & \infty \\ 50 & 0 & 15 & 5 \\ 30 & \infty & 0 & 15 \\ 15 & \infty & 5 & 0 \end{bmatrix} \end{matrix}$$

D_1, D_2, D_3, D_4 ?

الگوریتم:

$$D_k[i, j] = \min (D_{k-1}[i, j], D_{k-1}[i, k] + D_{k-1}[k, j])$$

محاسبه D_1 :

- $D_1[1, 1] = \min(D_0[1, 1], D_0[1, 1] + D_0[1, 1]) = \min(0, 0) = 0$
- $D_1[1, 2] = \min(D_0[1, 2], D_0[1, 1] + D_0[1, 2]) = \min(5, 5) = 5$
- $D_1[1, 3] = \min(D_0[1, 3], D_0[1, 1] + D_0[1, 3]) = \min(\infty, \infty) = \infty$
- $D_1[1, 4] = \min(D_0[1, 4], D_0[1, 1] + D_0[1, 4]) = \min(\infty, \infty) = \infty$
- $D_1[2, 1] = \min(D_0[2, 1], D_0[2, 1] + D_0[1, 1]) = \min(50, 50) = 50$
- $D_1[2, 2] = \min(D_0[2, 2], D_0[2, 1] + D_0[1, 2]) = \min(0, 55) = 0$
- $D_1[2, 3] = \min(D_0[2, 3], D_0[2, 1] + D_0[1, 3]) = \min(15, \infty) = 15$
- $D_1[2, 4] = \min(D_0[2, 4], D_0[2, 1] + D_0[1, 4]) = \min(5, \infty) = 5$
- $D_1[3, 1] = \min(D_0[3, 1], D_0[3, 1] + D_0[1, 1]) = \min(30, 30) = 30$
- $D_1[3, 2] = \min(D_0[3, 2], D_0[3, 1] + D_0[1, 2]) = \min(\infty, 35) = 35$
- $D_1[3, 3] = \min(D_0[3, 3], D_0[3, 1] + D_0[1, 3]) = \min(0, \infty) = 0$
- $D_1[3, 4] = \min(D_0[3, 4], D_0[3, 1] + D_0[1, 4]) = \min(15, \infty) = 15$
- $D_1[4, 1] = \min(D_0[4, 1], D_0[4, 1] + D_0[1, 1]) = \min(15, 15) = 15$
- $D_1[4, 2] = \min(D_0[4, 2], D_0[4, 1] + D_0[1, 2]) = \min(\infty, 20) = 20$
- $D_1[4, 3] = \min(D_0[4, 3], D_0[4, 1] + D_0[1, 3]) = \min(5, \infty) = 5$
- $D_1[4, 4] = \min(D_0[4, 4], D_0[4, 1] + D_0[1, 4]) = \min(0, \infty) = 0$

$$D_1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 5 & \infty & \infty \\ 50 & 0 & 15 & 5 \\ 30 & 35 & 0 & 15 \\ 15 & 20 & 5 & 0 \end{bmatrix} \end{matrix}$$

در ادامه محاسبه D_2

$$\begin{aligned}
 [1,1] &= \infty + D_1[2,1] & [2,1] &= \infty + D_1[2,1] & [3,1] &= \infty + D_1[2,1] \\
 [1,2] &= \infty + D_1[2,2] & [2,2] &= \infty + D_1[2,2] & [3,2] &= \infty + D_1[2,2] \\
 [1,3] &= \infty + D_1[2,3] & [2,3] &= \infty + D_1[2,3] & [3,3] &= \infty + D_1[2,3] \\
 [1,4] &= \infty + D_1[2,4] & [2,4] &= \infty + D_1[2,4] & [3,4] &= \infty + D_1[2,4] \\
 [4,1] &= \infty + D_1[2,1] & [4,2] &= \infty + D_1[2,2] & [4,3] &= \infty + D_1[2,3]
 \end{aligned}$$

$$D_2[1,1] = \min(D_1[1,1], D_4[1,2]) = \min(0, 55) = 0 \quad [4,4] = \infty + D_1[2,4]$$

$$D_2[1,2] = \min(D_1[1,2], D_4[1,2]) = \min(5, 5) = 5$$

$$D_2[1,3] = \min(D_1[1,3], D_4[1,2]) = \min(\infty, 20) = 20$$

$$D_2[1,4] = \min(D_1[1,4], D_4[1,2]) = \min(\infty, 10) = 10$$

$$D_2[2,1] = \min(D_1[2,1], D_4[2,2]) = \min(0, 50) = 50$$

$$D_2[2,2] = \min(D_1[2,2], D_4[2,2]) = \min(0, 0) = 0$$

$$D_2[2,3] = \min(D_1[2,3], D_4[2,2]) = \min(15, 15) = 15$$

$$D_2[2,4] = \min(D_1[2,4], D_4[2,2]) = \min(5, 5) = 5$$

$$D_2[3,1] = \min(D_1[3,1], D_4[3,2]) = \min(30, 85) = 30$$

$$D_2[3,2] = \min(D_1[3,2], D_4[3,2]) = \min(35, 35) = 35$$

$$D_2[3,3] = \min(D_1[3,3], D_4[3,2]) = \min(0, 50) = 0$$

$$D_2[3,4] = \min(D_1[3,4], D_4[3,2]) = \min(15, 40) = 15$$

$$D_2[4,1] = \min(D_1[4,1], D_4[4,2]) = \min(15, 70) = 15$$

$$D_2[4,2] = \min(D_1[4,2], D_4[4,2]) = \min(20, 20) = 20$$

$$D_2[4,3] = \min(D_1[4,3], D_4[4,2]) = \min(5, 35) = 5$$

$$D_2[4,4] = \min(D_1[4,4], D_4[4,2]) = \min(0, 25) = 0$$

$$D_2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 5 & 20 & 10 \\ 50 & 0 & 15 & 5 \\ 30 & 35 & 0 & 15 \\ 15 & 20 & 5 & 0 \end{bmatrix} \end{matrix}$$

← D_3 new (20, 10, 15, 5)

$$D_3[1,1] = \min(D_2[1,1], D_2[1,3] + D_2[3,1]) = \min(0, 50) = 0$$

$$D_3[1,2] = \min(D_2[1,2], D_2[1,3] + D_2[3,2]) = \min(5, 55) = 5$$

$$D_3[1,3] = \min(D_2[1,3], D_2[1,3] + D_2[3,3]) = \min(20, 20) = 20$$

$$D_3[1,4] = \min(D_2[1,4], D_2[1,3] + D_2[3,4]) = \min(10, 35) = 10$$

$$D_3[2,1] = \min(D_2[2,1], D_2[2,3] + D_2[3,1]) = \min(50, 45) = 45$$

$$D_3[2,2] = \min(D_2[2,2], D_2[2,3] + D_2[3,2]) = \min(0, 50) = 0$$

$$D_3[2,3] = \min(D_2[2,3], D_2[2,3] + D_2[3,3]) = \min(15, 15) = 15$$

$$D_3[2,4] = \min(D_2[2,4], D_2[2,3] + D_2[3,4]) = \min(5, 30) = 5$$

$$D_3[3,1] = \min(D_2[3,1], D_2[3,3] + D_2[3,1]) = \min(30, 30) = 30$$

$$D_3[3,2] = \min(D_2[3,2], D_2[3,3] + D_2[3,2]) = \min(35, 35) = 35$$

$$D_3[3,3] = \min(D_2[3,3], D_2[3,3] + D_2[3,3]) = \min(0, 0) = 0$$

$$D_3[3,4] = \min(D_2[3,4], D_2[3,3] + D_2[3,4]) = \min(15, 15) = 15$$

$$D_3[4,1] = \min(D_2[4,1], D_2[4,3] + D_2[3,1]) = \min(15, 45) = 15$$

$$D_3[4,2] = \min(D_2[4,2], D_2[4,3] + D_2[3,2]) = \min(20, 40) = 20$$

$$D_3[4,3] = \min(D_2[4,3], D_2[4,3] + D_2[3,3]) = \min(5, 5) = 5$$

$$D_3[4,4] = \min(D_2[4,4], D_2[4,3] + D_2[3,4]) = \min(0, 20) = 0$$

$$\begin{aligned}
 D_3[4,1] &= \min(D_2[4,1], D_2[4,3] + D_2[3,1]) = \min(15, 35) = 15 \\
 D_3[4,2] &= \min(D_2[4,2], D_2[4,3] + D_2[3,2]) = \min(20, 40) = 20 \\
 D_3[4,3] &= \min(D_2[4,3], D_2[4,3] + D_2[3,3]) = \min(5, 5) = 5 \\
 D_3[4,4] &= \min(D_2[4,4], D_2[4,3] + D_2[3,4]) = \min(0, 20) = 0
 \end{aligned}$$

$$D_3 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 5 & 20 & 10 \\ 45 & 0 & 15 & 5 \\ 30 & 35 & 0 & 15 \\ 15 & 20 & 5 & 0 \end{bmatrix} \end{matrix}$$

← D4 در ادامه محاسبه

$$\begin{aligned}
 D_4[1,1] &= \min(D_3[1,1], D_3[1,4] + D_3[4,1]) = \min(0, 25) = 0 \\
 D_4[1,2] &= \min(D_3[1,2], D_3[1,4] + D_3[4,2]) = \min(5, 30) = 5 \\
 D_4[1,3] &= \min(D_3[1,3], D_3[1,4] + D_3[4,3]) = \min(20, 15) = 15 \\
 D_4[1,4] &= \min(D_3[1,4], D_3[1,4] + D_3[4,4]) = \min(10, 10) = 10 \\
 D_4[2,1] &= \min(D_3[2,1], D_3[2,4] + D_3[4,1]) = \min(45, 20) = 20 \\
 D_4[2,2] &= \min(D_3[2,2], D_3[2,4] + D_3[4,2]) = \min(0, 25) = 0 \\
 D_4[2,3] &= \min(D_3[2,3], D_3[2,4] + D_3[4,3]) = \min(15, 10) = 10 \\
 D_4[2,4] &= \min(D_3[2,4], D_3[2,4] + D_3[4,4]) = \min(5, 5) = 5 \\
 D_4[3,1] &= \min(D_3[3,1], D_3[3,4] + D_3[4,1]) = \min(30, 30) = 30 \\
 D_4[3,2] &= \min(D_3[3,2], D_3[3,4] + D_3[4,2]) = \min(35, 35) = 35 \\
 D_4[3,3] &= \min(D_3[3,3], D_3[3,4] + D_3[4,3]) = \min(0, 20) = 0 \\
 D_4[3,4] &= \min(D_3[3,4], D_3[3,4] + D_3[4,4]) = \min(15, 15) = 15 \\
 D_4[4,1] &= \min(D_3[4,1], D_3[4,4] + D_3[4,1]) = \min(15, 15) = 15 \\
 D_4[4,2] &= \min(D_3[4,2], D_3[4,4] + D_3[4,2]) = \min(20, 20) = 20 \\
 D_4[4,3] &= \min(D_3[4,3], D_3[4,4] + D_3[4,3]) = \min(5, 5) = 5 \\
 D_4[4,4] &= \min(D_3[4,4], D_3[4,4] + D_3[4,4]) = \min(0, 0) = 0
 \end{aligned}$$

$$D_4 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 5 & 15 & 10 \\ 20 & 0 & 10 & 5 \\ 30 & 35 & 0 & 15 \\ 15 & 20 & 5 & 0 \end{bmatrix} \end{matrix}$$